

ECE 447

Lesson 04

Signals and Signal Space



UNITED STATES
AIR FORCE
ACADEMY



Life/Leadership Lesson of the Day

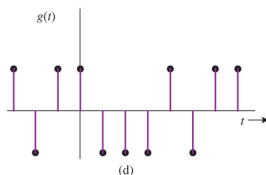
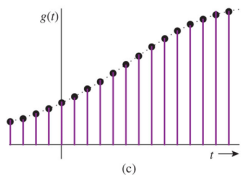
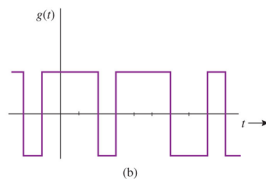
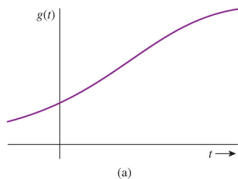


- **Lead!**
 - When you believe you can do it better than anyone else – step up!
- **Example**
 - Qatar built several large, nice dorms at AUAB
 - Internet to each room responsibility of 379th CS
 - No internet = low morale!
 - Capt Josh Thomas (in the CC's Support Staff) had the expertise; got permission from Wg/CC, organized cable patching teams
 - Completed within a few days
 - Fast internet = high morale!

SCHEDULE AND ADMIN

- [Schedule](#)
- Admin
 - **SDR Setup.** If unable to get FM radio station playing through GQRX last lesson - get help!
 - **Lab 1 Assignment.** The assignment associated with Lab 1 is due Lesson 6 - specifically 21 Aug by 2359 via Gradescope upload. Make sure you submit a **narrowband FM** signal - not a wideband stereo FM radio signal.
 - **Future HW problems...**

CLASSIFICATION OF SIGNALS

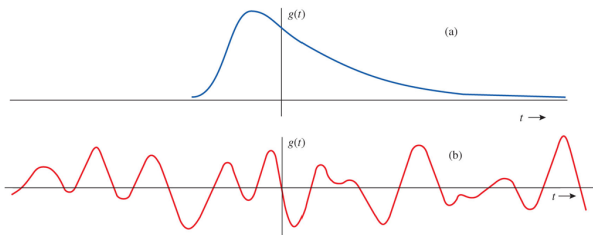


- Periodic vs. Aperiodic
- Deterministic vs. Random

SIGNAL SIZE

- Signal Energy: $E_g = \int_{-\infty}^{+\infty} |x(t)|^2 dt = \int_{-\infty}^{+\infty} x(t)x^*(t)dt$
- Signal Power:

$$E_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)x^*(t)dt$$



NORMALIZED RESISTANCE

We typically assume that $R = 1 \Omega$, i.e. *normalized*, so that

$$p(t) = |v(t)|^2 = |i(t)|^2 = x(t)^2 \text{ (Watts)}$$

Energy simply becomes

$$E = \int_{t_1}^{t_2} |x(t)|^2 dt \text{ (Joules)}$$

NORMALIZED RESISTANCE (CONT.)

Average power in the interval t_1 to t_2 is given by:

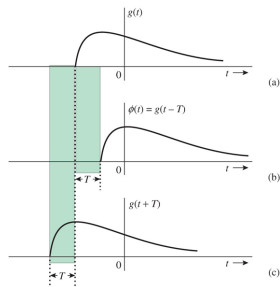
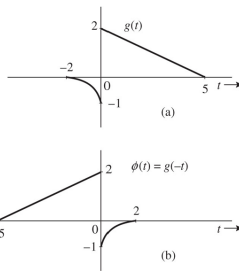
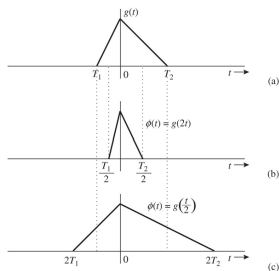
$$P_{avg} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} |x(t)|^2 dt$$

If the signal is periodic, $x(t + T) = x(t) \forall t$, then

$$P_{avg} = \frac{1}{T} \int_T |x(t)|^2 dt$$

USEFUL SIGNAL OPERATIONS

Name that operation!



IMPORTANT SIGNALS

- Unit Impulse:

$$\delta(t) = 0, t \neq 0 \quad \int_{-\infty}^{\infty} \delta(t) dt = 1 \quad \phi(t)\delta(t - T) = \phi(T)\delta(t - T)$$

- Unit Step:

$$\int_{-\infty}^t \delta(\tau) d\tau = u(t) =$$

BASIS FUNCTIONS

- Vector Analogy: two vectors are perpendicular or "orthogonal" when their dot product is zero

$$A_x \hat{x} \cdot A_y \hat{y} = 0$$

- They have nothing in common
- Cannot write an estimate of one in terms of the other
- Two signals are orthogonal on interval (t_1, t_2) if

$$\int_{t_1}^{t_2} x(t)y^*(t)dt = 0$$

BASIS FUNCTION EXAMPLE

- Consider functions $\sin(n\pi t)$ and $\sin(m\pi t)$ on interval $(0, 2)$
- Trig identity: $\sin(A)\sin(B) = \frac{1}{2}\cos(A-B) - \frac{1}{2}\cos(A+B)$

$$\int_0^2 \sin(n\pi t) \sin(m\pi t) dt = \frac{1}{2} \int_0^2 \cos[(n-m)\pi t] - \cos[(n+m)\pi t] dt$$

$$= \frac{1}{2\pi(n-m)} \sin[2\pi(n-m)t] - \frac{1}{2\pi(n+m)} \sin[2\pi(n+m)t] \Big|_0^2$$

$$= 0 \text{ for } n \neq m$$

BASIS FUNCTION EXAMPLE (CONT.)

- Consider functions $\sin(n\pi t)$ and $\sin(m\pi t)$ on interval $(0, 2)$
- Trig identity: $\sin(A) \sin(A) = \frac{1}{2} [1 - \cos(2A)]$

$$\begin{aligned}\int_0^2 \sin(n\pi t) \sin(n\pi t) dt &= \frac{1}{2} \int_0^2 1 - \cos(2n\pi t) dt \\&= 1 - \left. \frac{1}{4\pi n} \sin(4\pi n t) \right|_0^2 \\&= 1 \text{ for } n = m\end{aligned}$$

BASIS FUNCTION EXAMPLE (CONT.)

- Therefore, $\sin(n\pi t)$ and $\sin(m\pi t)$ are orthogonal from $(0, 2)$
- Because inner product = 1, they are orthonormal
- Mathematically, $\sin(n\pi t)$ and $\sin(m\pi t)$ have nothing in common
- Because there are an infinite number of n 's and m 's, we have an infinite number of basis functions
- *Do these functions span the space of all functions on $(0, 2)$?*

COMPLEX SIGNAL MODELS

- Rotating Phasor Representation:

$$\tilde{x}(t) = Ae^{j\theta}e^{j\omega_0 t} = Ae^{j(\omega_0 t + \theta)} = A\cos(\omega_0 t + \theta) + jA\sin(\omega_0 t + \theta)$$

- For orthogonality,

$$\int_{t_1}^{t_2} x_m(t)x_n^*(t)dt = 0 \quad m \neq n$$

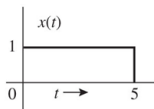
$$\int_{t_1}^{t_2} x_m(t)x_n^*(t)dt = E_n \quad m = n$$

- If $E_n = 1$, then signal set is *orthonormal*

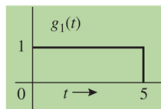
CORRELATION OF SIGNALS

- Generalized correlation definition:

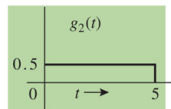
$$\rho = \frac{1}{\sqrt{E_g E_x}} \int_{-\infty}^{\infty} g(t) x^*(t) dt$$



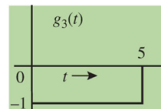
(a)



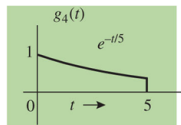
(b)



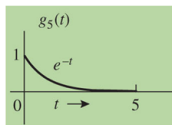
(c)



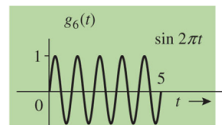
(d)



(e)



(f)



(g)

FOURIER SERIES REVIEW

- Synthesis
$$x(t) = \sum_{n=-N}^N X_n e^{jn\omega_0 t}$$

- Analysis
$$X_n = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} x(t) e^{-jn\omega_0 t} dt, \quad \omega_0 = \frac{2\pi}{T} = 2\pi f_0$$

- Average Power
$$P_{avg} = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} |x(t)|^2 dt = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) x^*(t) dt$$

$$P_{avg} = \sum_{n=-\infty}^{+\infty} X_n X_n^* = \sum_{n=-\infty}^{+\infty} |X_n|^2$$